

Integrated Design of a Controller and Compliant Power Supply for USB Power Delivery

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Abstract—This paper presents a design of an integrated controller and compliant power supply for USB Power Delivery. The design objectives focus on efficiency, cost-effectiveness, and reliability, while adhering to the requirements of USB Power Delivery. The system architecture comprises a power delivery controller and a power supply, designed using the RTL to GDS method and mixed-signal design respectively.

Index Terms—USB Power Delivery, AMS, BMC, SPR, EPR, BCD, Powercell

I. INTRODUCTION

UNIVERSAL Serial Bus (USB) has become the ubiquitous connector for data transfer and low-power device charging. However, traditional USB standards limited power delivery to 2.5W (USB 1.1) and 4.5W (USB 3.0). This restricted the use of USB for charging power-hungry devices, such as laptops and tablets. To address this limitation, USB Implementers Forum (USB-IF) introduced the USB Power Delivery (USB PD) specification.

USB PD standard was introduced in 2012 alongside the USB-C connector. It allows devices to transfer higher levels of power over a USB connection, significantly speeding up the charging process. USB PD can deliver up to 100 watts of power in the Standard Power Range (SPR), enough to charge not only smartphones and tablets but also larger devices like laptops. The latest version, USB PD 3.1, extends the power capability even further, supporting up to 240 watts. This extended capability is termed as Extended Power Range (EPR).

II. NEED FOR INTEGRATED CONTROLLERS AND COMPLIANT POWER SUPPLIES

The evolution of USB PD has increased the complexity of designing power supplies. Integrated controllers and compliant power supplies are essential for several reasons:

- **Efficient Power Management:** Integrated controllers help manage power delivery efficiently across multiple peripherals, ensuring each device gets the power it needs.
- **Safety and Compliance:** Compliant power supplies ensure that devices are charged safely and meet industry standards, reducing the risk of damage or malfunction.
- **Cost-Effectiveness:** Using integrated controllers can reduce the overall cost of the Bill-of-Materials (BOM) by combining multiple functions into a single chip.
- **Design Flexibility:** Integrated controllers offer flexibility in design, allowing for custom power supply circuits and unique charging algorithms.

By leveraging these integrated solutions, designers can create USB power supplies that are not only efficient and safe, but also capable of meeting the growing power demands of modern devices.

III. EXISTING SOLUTIONS AND CHALLENGES

Major technology companies such as Texas Instruments, Analog Devices, Infineon, and others have developed USB PD solutions, typically in the form of separate chips for the controller, gate driver, current regulator, and Power FET. Although these solutions have proven to be effective, they also occupy a significant footprint on the device, which can be a limiting factor in the design and miniaturization of electronic devices.

The concept of integrating the controller, gate driver, current regulator, and Power FET into a single chip presents an intriguing opportunity to save footprint and potentially improve reliability. This approach could streamline the power delivery process, reduce the complexity of the device design, and allow for more efficient use of space. However, the integration of these components into a single chip is not without its challenges, and these challenges form the focus of the present work.

The present work is an exploration of the concept of an integrated controller for USB PD solutions. It aims to understand the challenges associated with this integration and to develop strategies to overcome them. By examining the performance of the integrated controller in comparison to the separate chip solutions currently in use, this work will provide valuable insights into the feasibility and potential benefits of this integrated approach.

IV. METHODOLOGY

- 1) **Designing the Circuit:** The first step is to design the RTL for PD controller and schematic entry for the power supply and its supporting circuitry.
- 2) **Pre-layout AMS Simulation:** After the schematic design and RTL is complete, the next step is to perform a pre-layout AMS simulation of both devices using Spectre AMS. This involves applying the appropriate input signals to the circuit and observing the output signals.
The simulation should verify that the circuit meets the desired performance metrics, such as output voltage, dropout voltage, current regulation, transient response, and stability.

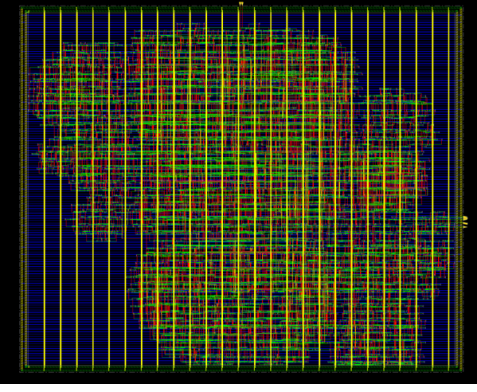


Fig. 1. Post-Route view of Controller

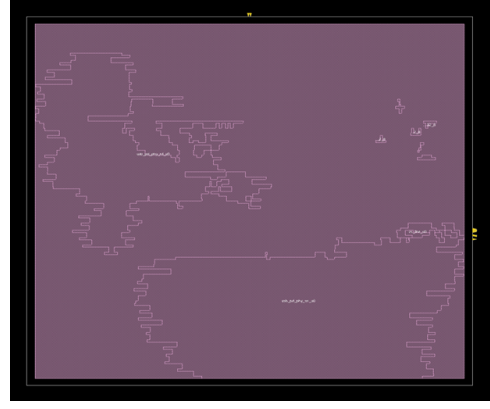


Fig. 2. Logical Partitioning of Controller

- 3) **Layout Generation:** After the pre-layout simulation confirms that the schematic design meets the desired performance metrics, the next step is to generate the layout of the power supply circuit using Cadence Virtuoso and controller through digital PNR flow. The layout should be designed in such a way that it minimizes the area of the circuit while maintaining the performance of the circuit. The layout should also take into account the design rules of the 45nm BCD (Bipolar-CMOS-DMOS) technology node.
- 4) **Parasitic Extraction:** Once the layout is complete, the next step is to extract the parasitics of the circuit using Quantus RC. This involves identifying the parasitic resistances, capacitances, and inductances in the circuit that can affect the performance of the circuit. The extracted parasitics are then back-annotated to the schematic.
- 5) **Post-layout AMS Simulation:** After the parasitics are back-annotated to the schematic, the next step is to perform a post-layout AMS simulation of the circuit using Spectre AMS. This involves applying the appropriate input signals to the circuit and observing the output signals. The simulation should verify that the circuit meets the desired performance metrics, taking into account the effects of the extracted parasitics.
- 6) **Performance Evaluation and Comparison:** The final step is to evaluate the performance of the circuit and compare it with existing designs. This involves analyzing the simulation results and determining whether the circuit meets the desired performance metrics.

V. PD CONTROLLER

The power delivery controller was meticulously designed using 1.2V, 9T RVT (Regular Voltage Threshold) standard cells, leveraging the advanced 45BCD (Bipolar-CMOS-DMOS) technology node. This choice of technology node is particularly advantageous for integrating high-voltage and high-current analog components with low-power digital circuits, making it ideal for mixed-signal applications. The design process employed an OA (OpenAccess) based flow, which significantly enhances the integration and interoperability of the design within the Cadence Virtuoso environment. OpenAccess

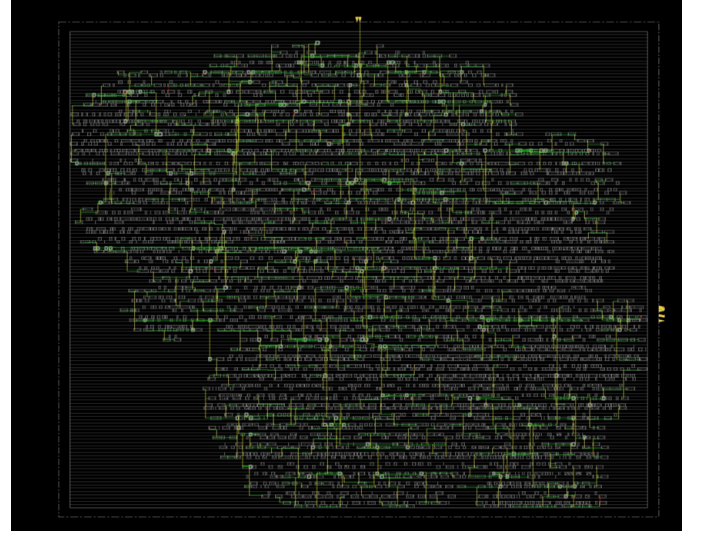


Fig. 3. CTS Implementation of Controller

is a widely adopted database standard that facilitates seamless data exchange and tool interoperability, thereby streamlining the design process and reducing the likelihood of errors.

The use of 9T RVT standard cells ensures a balanced trade-off between performance and power consumption, which is critical for power delivery applications where efficiency and reliability are paramount. These cells are optimized for low leakage and high drive strength, contributing to the overall robustness of the power delivery controller. The 1.2V operating voltage is chosen to align with industry standards for low-power design, ensuring compatibility with a wide range of digital and analog components.

In the Cadence Virtuoso environment, the OA-based flow allows for efficient co-design and co-simulation of analog and digital circuits. This is particularly beneficial for mixed-signal designs, where the interaction between analog and digital components must be carefully managed to ensure overall system performance. The integration of the power delivery controller with other analog and mixed-signal circuits is facilitated by the comprehensive suite of tools available in Virtuoso, including schematic capture, layout, and verification tools. This holistic approach ensures that the design meets all

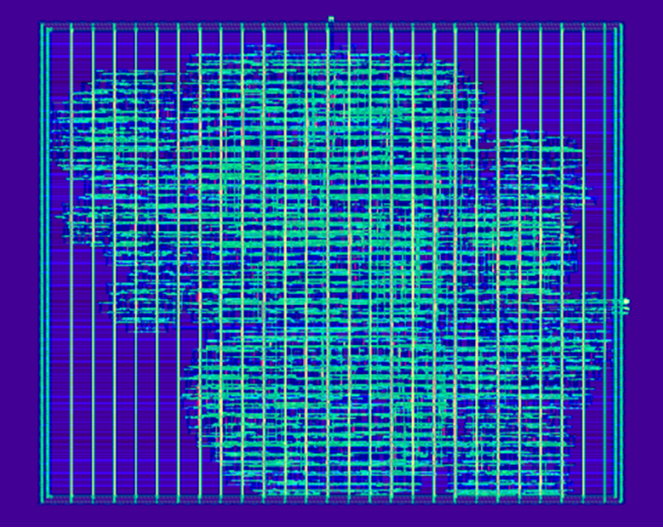


Fig. 4. EMIR view of PD Controller

performance, power, and area requirements while maintaining high levels of reliability and manufacturability. Overall, the combination of 45BCD technology, 9T RVT standard cells, and OA-based design flow in Cadence Virtuoso provides a robust framework for developing high-performance power delivery controllers. Electromigration and IR Drop (EMIR) analysis reveals hotspots in a design, it indicates areas where current density is too high or voltage drops are significant, potentially leading to reliability issues. Tackling these hotspots effectively is crucial for ensuring the longevity and performance of the power delivery controller. Utilizing Low Voltage Threshold (LVT) and High Voltage Threshold (HVT) cells can be a strategic approach to mitigate these issues. LVT cells, with their lower threshold voltages, switch faster and handle higher currents, making them ideal for critical paths where performance is essential. This localized enhancement can help distribute the current more evenly, reducing stress on any single point and mitigating hotspots. Conversely, HVT cells, with higher threshold voltages, have lower leakage currents, making them suitable for non-critical paths to minimize power consumption and heat generation. This hybrid approach balances performance and power efficiency, ensuring optimal thermal management. Additionally, layout optimization, such as redesigning power grids and adding decoupling capacitors, can further enhance current distribution and voltage stability, addressing the hotspots identified during EMIR analysis and ensuring a robust and reliable power delivery controller.

TABLE I
SPECIFICATIONS AND RESULTS OF POWER DELIVERY CONTROLLER

Power Delivery Controller		
Parameter	Specification	Result
Nominal Voltage	1.2V	1.2V
Total Power Consumption	<1mW	0.5114 mW
Std. Cell Count	<10K	5502
Clock Frequency	100MHz	100MHz
Die Area	<1mm ²	0.7563 mm ²

VI. POWER SUPPLY

The power supply design integrates three key components: a Buck Converter, an LDO (Low Dropout Regulator), and a Current Regulator. Ultra-low $R_{ds(on)}$ (ULR) PMOS transistors are employed as the switching elements in the Buck Converter and as pass elements in both the LDO and the Current Regulator. These PMOS transistors are configured in a power cell arrangement, which is advantageous for handling high currents and ensuring efficient power delivery. However, due to their large size and high gate capacitance, suitable driver circuits are necessary to drive these transistors effectively.

The Buck Converter utilizes a 2-bit V_{sel} signal to select the appropriate driver RC (resistor-capacitor) network for PWM (Pulse Width Modulation) generation. This selection mechanism ensures that the driver circuit can adapt to different operating conditions, optimizing the performance of the Buck Converter. The PWM signal operates at a frequency of 40 kHz, which is chosen to balance efficiency and response time. By varying the duty cycle of the PWM signal, the output voltage of the Buck Converter can be precisely controlled to meet the desired specifications.

In the LDO and Current Regulator, the PMOS transistors serve as pass elements, regulating the output voltage and current to ensure stable and reliable power delivery. The use of ULR PMOS transistors minimizes power losses and enhances the overall efficiency of the power supply. The integration of these components into a cohesive power supply design ensures that the system can deliver the required power with high efficiency and reliability, making it suitable for a wide range of applications.

As indicated by the Table II, the design necessitates a relatively large inductance to store energy and ensure continuous current flow to the output. This is crucial for maintaining a stable output voltage and minimizing disruptions in power delivery. However, the output of the Buck Converter inherently contains ripples due to the switching nature of the converter. To mitigate these ripples and achieve a smooth DC output, a capacitor is employed to filter out the high-frequency components. The chosen capacitor must be reliable and capable of withstanding voltage spikes, which are common in switching power supplies.

Given the substantial size of the required inductor and capacitor, integrating these components on-chip is impractical. On-chip inductors and capacitors would occupy excessive silicon area and may not provide the necessary performance characteristics. Therefore, the design incorporates pads to facilitate the external connection of these components. By using external components, the design can leverage high-quality, high-performance inductors and tantalum capacitors, which are well-suited for handling the energy storage and filtering requirements of the Buck Converter.

Tantalum capacitors are particularly favored for their stability, reliability, and tolerance to voltage spikes, making them ideal for filtering applications in power supplies. The external connection of the inductor and capacitor allows for greater flexibility in component selection and optimization, ensuring that the power supply meets the stringent performance and

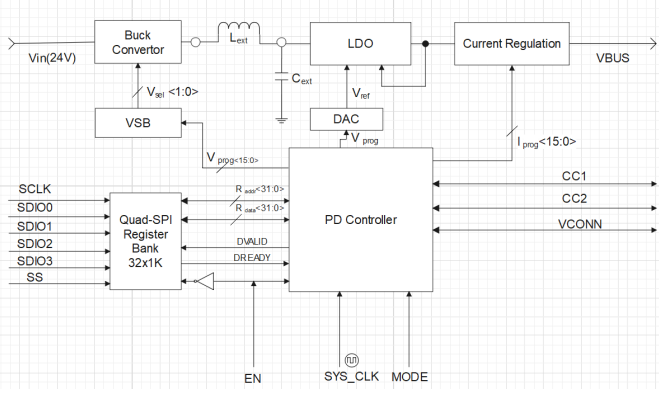


Fig. 5. Block Diagram of Integrated System

reliability criteria. This approach balances the need for high-performance passive components with the practical constraints of on-chip integration, resulting in a robust and efficient power supply design.

TABLE II
SPECIFICATIONS AND RESULTS OF PROGRAMMABLE POWER SUPPLY

Programmable Power Supply		
Parameter	Specification	Result
Operating Input Voltage	24V	24V
Maximum Output Power	65W	65W
Total Dissipated Power	<2W	5.0025W
Switching frequency	40KHz	40KHz
Die Area	<25mm ²	4.3122
Max. Regulated Current O/P	5A	5.2A
Total Consumed Power	-	68.3W

VII. RESULTS

The results of the power delivery controller and programmable power supply are presented in a comparative table, highlighting key parameters and their corresponding specifications and outcomes. For the power delivery controller, the nominal voltage was specified at 1.2V, which was accurately achieved. The total power consumption was targeted to be less than 1mW, and the actual result was 0.5114mW, indicating efficient power usage. The standard cell count was specified to be under 10K, with the actual count being 5502 cells, demonstrating a compact design. The clock frequency met the specified 100MHz, and the die area was slightly larger than the target of 1mm², at 0.7563mm².

In contrast, the programmable power supply had an operating input voltage of 24V, which was met precisely. The maximum output power was specified and achieved at 65W. However, the total dissipated power exceeded the target of 2W, resulting in 5.0025W, suggesting potential inefficiencies. The switching frequency was accurately maintained at 40kHz. The die area was significantly compact than the specified 25mm², at 4.3122mm², excluding Power MOSFETs. The maximum regulated current output was not specified but resulted in 5.2A. The resulting fully integrated chip has a dimension of 12.75mm x 12.875mm. These results provide a comprehensive overview of the performance and areas for improvement in the design.

VIII. CONCLUSION

In conclusion, the integration of the controller, gate driver, current regulator, and Power FET into a single chip for USB PD solutions represents a promising direction for future research. While the challenges associated with this integration are significant, the potential benefits in terms of footprint reduction and improved reliability make it a worthwhile endeavor. The findings of the present work will contribute to the ongoing discourse in this field and may pave the way for more efficient and compact USB PD solutions in the future.

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